

# RFM Customer Segmentation & Retention Analytics

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## **Complete SQL Methodology & Business Logic Documentation**

MySQL 8.0 • Power BI • RFM Analysis • Retention Strategy

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This document explains the SQL pipeline contained in RFM\_Analysis\_Final.sql, including data preparation, RFM calculations, scoring, segmentation, risk classification, executive KPIs, and the strategic marketing investment simulation.

## 1. Project Overview

The project analyzes the UCI Online Retail dataset at customer level. The objective is to transform raw transactional data into actionable customer segments, identify customers showing recency-based risk, quantify revenue at risk, and simulate retention investment priorities.

| Metric                  | Value                  |
|-------------------------|------------------------|
| Raw transactions        | 541,909                |
| Valid transactions      | 397,880                |
| Customers               | 4,338                  |
| Countries               | 38                     |
| Total revenue           | £8,911,407.90          |
| Champions               | 558                    |
| Champions revenue share | 56.87%                 |
| High-risk customers     | 1,449                  |
| Revenue at Risk         | £1,035,270.41 (11.62%) |
| Portfolio churn risk    | 6.22 — Warning         |

## 2. Analytical Architecture

The SQL implementation follows a layered architecture: Bronze → Silver → Core Metrics → Gold → BI → Strategic.

```
RAW DATA
↓
BRONZE: transactions
↓
SILVER: fact_transactions
↓
CORE: rfm_base → rfm_scores → rfm_scores_with_composite
↓
GOLD: gold_customer_insights
↓
BI: segment_summary + business_kpis
↓
STRATEGIC: strategic_customer_actions
```

Each view has a defined business purpose and is followed by validation queries in the SQL script.

## 3. Dataset and Data Ingestion

Source: UCI Online Retail Dataset distributed through Kaggle. The dataset covers December 2010 through December 2011 and contains 541,909 raw rows.

The transactions table is designed as the raw Bronze-layer table.

```
CREATE TABLE transactions (
  id_transactions INT NOT NULL AUTO_INCREMENT,
  InvoiceNo VARCHAR(45) NULL,
  StockCode VARCHAR(45) NULL,
  Description VARCHAR(255) NULL,
  Quantity INT NULL,
  InvoiceDate VARCHAR(50) NULL,
  UnitPrice DECIMAL(10,2) NULL,
  CustomerID INT NULL,
  Country VARCHAR(100) NULL,
```

```
PRIMARY KEY (id_transactions)
);
```

The script provides two LOAD DATA INFILE options: a custom folder requiring secure\_file\_priv configuration and the default MySQL Uploads folder.

## 4. Section 2 — Data Preparation & Cleaning

The source InvoiceDate field is stored as VARCHAR. A clean DATETIME column is populated with STR\_TO\_DATE. Country values are also cleaned of Windows carriage-return characters.

```
UPDATE transactions
SET InvoiceDate_clean = STR_TO_DATE(InvoiceDate, '%m/%d/%Y %H:%i')
WHERE InvoiceDate IS NOT NULL
  AND InvoiceDate != '';

UPDATE transactions
SET Country = TRIM(REPLACE(Country, '\r', ''))
WHERE Country LIKE '%\r%';
```

Validation checks confirm date population and the number of distinct countries.

## 5. Section 3 — fact\_transactions (Silver Layer)

fact\_transactions is the clean analytical transaction view. It removes records that cannot be used for positive-sales RFM analysis.

```
CREATE OR REPLACE VIEW fact_transactions AS
SELECT
  InvoiceNo,
  StockCode,
  Description,
  Quantity,
  InvoiceDate_clean AS InvoiceDate,
  UnitPrice,
  CustomerID,
  Country,
  ROUND(Quantity * UnitPrice, 2) AS total_amount
FROM transactions
WHERE CustomerID IS NOT NULL
  AND Quantity > 0
  AND UnitPrice > 0
  AND InvoiceDate_clean IS NOT NULL;
```

- NULL CustomerID rows are excluded because they cannot be attributed to a customer.
- Negative or zero quantities are excluded from the positive-sales analysis.
- Zero or negative unit prices are excluded.
- Rows without a valid cleaned date are excluded.
- Expected analytical volume: approximately 397,880 valid rows.

## 6. Section 4 — RFM Base Metrics

RFM metrics are calculated per customer using the maximum invoice date in the cleaned dataset as the reference date.

| Metric    | SQL logic                                     | Interpretation           |
|-----------|-----------------------------------------------|--------------------------|
| Recency   | DATEDIFF(MAX dataset date, MAX customer date) | Days since last purchase |
| Frequency | COUNT(DISTINCT InvoiceNo)                     | Number of unique orders  |
| Monetary  | SUM(total_amount)                             | Total customer spend     |

```
WITH reference_context AS (  
    SELECT MAX(InvoiceDate) AS max_invoice_date  
    FROM fact_transactions  
)  
,  
customer_metrics AS (  
    SELECT  
        CustomerID,  
        MAX(InvoiceDate) AS last_purchase_date,  
        DATEDIFF(  
            (SELECT max_invoice_date FROM reference_context),  
            MAX(InvoiceDate)  
        ) AS Recency,  
        COUNT(DISTINCT InvoiceNo) AS Frequency,  
        ROUND(SUM(total_amount), 2) AS Monetary  
    FROM fact_transactions  
    GROUP BY CustomerID  
)  
SELECT * FROM customer_metrics;
```

Using the dataset's own maximum transaction date makes Recency dynamic to the data rather than dependent on the current calendar date.

## 7. Section 5 — Fixed RFM Scoring

The project uses fixed thresholds rather than percentile or NTILE scoring. This makes the business rules easy to explain and stable when the dataset distribution changes.

| Score         | Recency     | Frequency   | Monetary     |
|---------------|-------------|-------------|--------------|
| 4 — Excellent | ≤ 30 days   | ≥ 10 orders | ≥ £5,000     |
| 3 — Good      | 31–90 days  | 5–9 orders  | £2,000–4,999 |
| 2 — Fair      | 91–180 days | 2–4 orders  | £500–1,999   |
| 1 — Poor      | > 180 days  | 1 order     | < £500       |

Recency is inverse-scored: fewer days since purchase receives a higher score. Frequency and Monetary are positively scored: higher activity and spend receive higher scores.

## 8. Composite RFM Score

$$\text{RFM Composite} = R \times 0.30 + F \times 0.35 + M \times 0.35$$

Frequency and Monetary each receive 35% weight, while Recency receives 30%. The resulting score is rounded to two decimals and labeled Excellent, Good, Fair, or Poor.

| Label     | Composite score |
|-----------|-----------------|
| Excellent | ≥ 3.5           |

|      |                        |
|------|------------------------|
| Good | $\geq 2.5$ and $< 3.5$ |
| Fair | $\geq 1.5$ and $< 2.5$ |
| Poor | $< 1.5$                |

## 9. Section 6 — Gold Customer Insights

gold\_customer\_insights converts the analytical scores into business dimensions: segment, value tier, risk group, marketing action, and churn risk.

### 9.1 Customer Segmentation

| Segment             | Business rule             |
|---------------------|---------------------------|
| Champions           | $R=4, F \geq 3, M \geq 3$ |
| Loyal Customers     | $F \geq 3, M \geq 3$      |
| New Customers       | $R=4, F \leq 2$           |
| Potential Loyalists | $R \geq 3, F=2$           |
| At Risk             | $R \leq 2, F \geq 3$      |
| Lost High Value     | $R=1, M \geq 3$           |
| Lost                | $R=1$                     |
| Needs Attention     | Remaining customers       |

CASE statement order matters. Earlier conditions take precedence when a customer could satisfy more than one conceptual category.

### 9.2 Value Tier

| Tier       | Monetary rule                 |
|------------|-------------------------------|
| High Value | $\geq \text{£}5,000$          |
| Mid Value  | $\text{£}2,000\text{--}4,999$ |
| Low Value  | $< \text{£}2,000$             |

### 9.3 Risk Group

```
CASE
  WHEN r_score <= 2 THEN 'High Risk'
  WHEN r_score = 3 THEN 'Medium Risk'
  ELSE 'Low Risk'
END
```

The High Risk definition is explicitly recency-driven: customers with  $r\_score \leq 2$  are High Risk regardless of segment.

### 9.4 Marketing Action

| Condition                 | Marketing action       |
|---------------------------|------------------------|
| $R=4$ and $F \geq 3$      | Retain & Upsell        |
| $R=4$ and $F < 3$         | Welcome & Nurture      |
| $R \leq 2$ and $F \geq 3$ | Re-engagement Campaign |
| $R=1$                     | Win-back Offer         |
| Otherwise                 | Monitor                |

### 9.5 Churn Risk Score

```
Churn Risk =
(5 - R_score) × 1.5
+ (5 - F_score) × 1.0
```

The theoretical score range is 2.5–10.0. Higher values indicate greater risk. The formula emphasizes Recency more strongly than Frequency.

## 10. Section 7 — BI / Executive Views

### 10.1 segment\_summary

segment\_summary aggregates customer count, revenue, average customer value, revenue share, cumulative revenue share, average behavior metrics, RFM score, churn risk, and segment health.

```
Segment health:
  Average churn risk >= 7.0   → Critical
  Average churn risk >= 4.5   → Warning
  Otherwise                   → Healthy
```

The cumulative revenue share supports Pareto-style analysis of which segments drive the largest proportion of revenue.

### 10.2 business\_kpis

business\_kpis produces a single-row executive summary used for dashboard KPI cards.

| KPI                     | Current result |
|-------------------------|----------------|
| Total customers         | 4,338          |
| Champion customers      | 558            |
| High-risk customers     | 1,449          |
| High-value customers    | 275            |
| Total revenue           | £8,911,407.90  |
| Average customer value  | £2,054.27      |
| Champions revenue       | £5,068,266.70  |
| Champions revenue share | 56.87%         |
| Revenue at Risk         | £1,035,270.41  |
| Revenue at Risk %       | 11.62%         |
| Portfolio RFM score     | 2.21           |
| Portfolio churn risk    | 6.22           |
| Portfolio health        | Warning        |

Revenue at Risk is intentionally aligned with the Power BI definition: total Monetary value for customers whose risk\_group is High Risk.

```
SUM(
  CASE
    WHEN risk_group = 'High Risk'
    THEN Monetary
    ELSE 0
  END
) AS revenue_at_risk
```

## 11. Revenue at Risk — Final Business Definition

Revenue at Risk is not defined by the segment names At Risk, Lost, or Lost High Value. Instead, it is defined by the independent risk\_group classification.

```
Revenue at Risk =
SUM(Monetary)
```

```
WHERE risk_group = 'High Risk'
```

```
High Risk =  
r_score <= 2
```

This broad definition captures customers whose recency has deteriorated even when their segment label has not yet moved into an explicitly negative segment.

Current value: £1,035,270.41, equal to 11.62% of total revenue.

## 12. Section 8 — Strategic Investment Simulation

strategic\_customer\_actions assigns an investment rate by customer segment and calculates simulated investment, estimated customer value, ROI ratio, recovery priority, and action urgency.

| Segment             | Investment rate |
|---------------------|-----------------|
| Champions           | 5%              |
| Loyal Customers     | 7%              |
| Potential Loyalists | 10%             |
| New Customers       | 12%             |
| Needs Attention     | 8%              |
| At Risk             | 18%             |
| Lost High Value     | 20%             |
| Lost                | 3%              |

These are scenario assumptions embedded in the SQL business logic. They are not observed campaign costs.

### 12.1 Investment Calculations

```
Estimated Marketing Investment  
= Monetary × investment_rate
```

```
Estimated Customer Value  
= Monetary × (1 - investment_rate)
```

```
ROI Ratio  
= (1 - investment_rate) / investment_rate
```

The ROI ratio is interpreted as simulated retained value generated per £1 of investment under the assumed scenario.

### 12.2 Recovery Priority Score

```
Recovery Priority Score =  
(churn_risk_score × 0.4)  
+  
(investment_rate × 10 × 0.6)
```

Higher values indicate higher priority under the project's combined risk/investment logic.

### 12.3 Action Urgency

| Condition           | Urgency           |
|---------------------|-------------------|
| Churn risk $\geq 8$ | Immediate Action  |
| Churn risk $\geq 6$ | Action This Month |
| Churn risk $\geq 4$ | Monitor Closely   |
| Otherwise           | Maintain          |

## 13. Retained Value vs. Recovery Potential

The strategic SQL view calculates Estimated Customer Value for each customer. The Power BI dashboard then aggregates this into a gross retained-value view and calculates a net Recovery Potential by subtracting the total simulated marketing investment.

| Metric                   | Formula                               | Value      |
|--------------------------|---------------------------------------|------------|
| Retained Value (Gross)   | SUM(Estimated Customer Value)         | £8,295,400 |
| Marketing Investment     | SUM(Estimated Marketing Investment)   | £616,009   |
| Recovery Potential (Net) | Retained Value – Marketing Investment | £7,679,391 |

Recovery Potential is therefore a scenario-based net value estimate, not observed historical revenue recovery.

## 14. Power BI Alignment

The Power BI dashboard uses the same customer-level gold layer and business definitions. The Revenue At Risk measure filters gold\_customer\_insights to risk\_group = 'High Risk'.

```
Revenue At Risk (High Risk Customers) =  
CALCULATE(  
    [Total Revenue],  
    'rfm_analysis gold_customer_insights'[risk_group] = "High Risk"  
)
```

The dashboard's Recovery Potential measure is:

```
Recovery Potential =  
[Estimated Customer Value]  
- [Estimated Marketing Investment]
```

This produces the net Recovery Potential shown on the dashboard.

## 15. Validation Checklist

- Raw row count: 541,909.
- Valid transaction row count: approximately 397,880.
- Customer count: 4,338.
- No NULL segments in gold\_customer\_insights.
- High-risk customers: 1,449.
- Revenue at Risk: £1,035,270.41.
- Revenue at Risk percentage: 11.62%.
- Champions: 558 customers.
- Champions revenue share: 56.87%.
- Portfolio churn risk: 6.22.
- Portfolio health: Warning.
- Strategic investment and retained-value totals should reconcile with the Power BI dashboard.



## 16. SQL Object Reference

| Section | Object                         | Purpose                             |
|---------|--------------------------------|-------------------------------------|
| 1       | transactions                   | Raw data ingestion                  |
| 2       | transactions.InvoiceDate_clean | Clean DATETIME conversion           |
| 3       | fact_transactions              | Valid analytical transactions       |
| 4       | rfm_base                       | Customer-level RFM metrics          |
| 5       | rfm_scores                     | Fixed-threshold R/F/M scoring       |
| 5       | rfm_scores_with_composite      | Weighted composite score            |
| 6       | gold_customer_insights         | Segments, value, risk, actions      |
| 7       | segment_summary                | Segment-level BI aggregation        |
| 7       | business_kpis                  | Executive KPI summary               |
| 8       | strategic_customer_actions     | Investment and retention simulation |

## 17. Repository and Reproducibility

The SQL file is designed to be executed in MySQL 8.0. The database dump provides a fast setup option, while the SQL script provides the full transformation logic and validation steps.

For a clean rebuild, follow the comments in Section 1 and Section 2 regarding CREATE TABLE, LOAD DATA INFILE, secure\_file\_priv, and the first-run InvoiceDate\_clean column.

## 18. Interpretation Notes

- RFM thresholds are business rules, not statistically optimized cutoffs.
- The churn-risk score is an analytical prioritization score, not a machine-learning probability of churn.
- Revenue at Risk is historical revenue associated with recency-based High Risk customers; it is not a forecast of future lost revenue.
- ROI and investment figures are simulated scenarios based on segment-level assumptions.
- Recovery Potential is a modeled net value estimate, not observed campaign revenue recovery.
- The project's purpose is decision support: identifying where retention effort may be most valuable.

## 19. Key Business Findings

- Champions generate 56.87% of total revenue while representing approximately 13% of customers.
- Needs Attention is the largest segment, making it an important population for proactive engagement.
- Lost High Value is small in customer count but material in historical revenue, making it a high-priority win-back population.
- Retain & Upsell has the strongest simulated ROI ratio under the project's investment assumptions.
- The portfolio-wide churn risk of 6.22 places the customer base in the Warning health zone.
- £1.04M of historical revenue is associated with recency-based High Risk customers.

## 20. Author

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